

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**Supplementary Winter Examination – 2025****Course: B.Tech****Branch : Mechanical Engineering****Semester :VIII****Subject Code & Name: BTMEC801A Fundamentals of Automotive Systems****Max Marks: 60****Date:****Duration: 3 Hr.****Instructions to the Students:**

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The ideal cycle for a petrol engine is.	CO1	1
	a. Diesel cycle b. Otto cycle c. Dual cycle d. Rankine cycle		
2	Which of the following has the highest thermal efficiency?	CO1	1
	a. Two-stroke petrol engine b. Four-stroke petrol engine c. Two-stroke diesel engine d. Four-stroke diesel engine		
3	The clearance volume of an IC engine is	CO1	1
	a. Volume displaced by piston b. Volume of cylinder c. Volume above piston at TDC d. Volume below piston at BDC		
4	The compression ratio of a CI engine is generally	CO2	1
	a. 4:1 to 6:1 b. 6:1 to 8:1 c. 8:1 to 10:1 d. 14:1 to 25:1		
5	The thermal efficiency of a CI engine is higher than SI engine because of	CO2	1
	a. Lower fuel consumption b. Higher compression ratio c. Better lubrication d. Less friction		
6	Exhaust Gas Recirculation (EGR) is mainly used to reduce	CO3	1
	a. CO emissions b. HC emissions c. NOx emissions d. SO ₂ emissions		
7	At which operating condition is Exhaust Gas Recirculation generally	CO3	1

	not used?					
	a. Idle condition	b. Medium load	c. Part load	d. Full load		
8	The purpose of clutch slipping during engagement is to				CO3	1
	a. Increase torque	b. Avoid sudden shock	c. Increase speed	d. Reduce friction		
9	Which type of gearbox is most commonly used in heavy trucks?				CO4	1
	a. Sliding mesh gearbox	b. Constant mesh gearbox	c. Synchromesh gearbox	d. Epicyclic gearbox		
10	The hydraulic pump in a power steering system is driven by				CO4	1
	a. Battery	b. Electric motor	c. Engine crankshaft via belt	d. Steering wheel		
11	Which suspension system is most commonly used in passenger cars?				CO5	1
	a. Rigid axle suspension	b. Leaf spring suspension	c. Air suspension	d. Independent suspension		
12	The MacPherson strut suspension consists of				CO5	1
	a. Leaf spring and axle	b. Coil spring and shock absorber combined	c. Torsion bar and damper	d. Air spring and compressor		
Q. 2	Solve the following.					12
A)	What is an engine block? Explain wet sleeve and dry sleeve inserts with neat sketches				CO1,CO2	6
B)	Explain the loop scavenging process in two-stroke engine with neat schematic diagram				CO1,CO2	6
Q.3	Solve the following.					12
A)	Explain different phases of Combustion in IC engine with neat sketch				CO1, CO2	6
B)	Explain exhaust gas recirculation (EGR) method for controlling the emissions from the engine.				CO1, CO2	6

Q. 4	Solve Any Two of the following.		12
A)	Explain the working of clutch in automotive with neat sketch.	CO3, CO4	6
B)	Explain the construction and working of synchromesh gear box with neat sketch.	CO3, CO4	6
C)	Explain the engine power-torque vs speed characteristics for actual IC engine with suitable performance curves	CO3, CO4	6
Q.5	Solve Any Two of the following.		12
A)	What is antilock braking system (ABS)? Explain working with neat sketch.	CO2, CO3	6
B)	Explain the construction and working of Drum Brake with neat sketch.	CO2, CO3	6
C)	Explain the working of Macpherson Strut Suspension system with neat sketch	CO4, CO5	6
Q. 6	Solve Any Two of the following.		12
A)	Explain the working of hydraulic power steering system with neat sketch	CO4, CO5	6
B)	Describe the various types of front suspension systems	CO4, CO5	6
C)	Differentiate between tyre with tube and tubeless tyre.	CO4, CO5	6
	*** End ***		